Seminar Report - Formatting Instructions (S7, 2026–27)

Department of Computer Science and Engineering,
Amal Jyothi College of Engineering (Autonomous), Kanjirappally

These instructions are derived directly from the LaTeX template (Report_format_S7.zip). Students preparing the report must follow them exactly.

1. Page Setup

Parameter	**Value**
Paper size	A4
Left margin (binding edge)	1.2 in / 3 cm
Right margin	1.0 in / 2.5 cm
Top margin	1.0 in / 2.5 cm
Bottom margin	1.0 in / 2.5 cm

2. Font and Spacing

Parameter	**Value**
Body font (LaTeX)	Computer Modern Roman (default report class font)
Body font (Word equivalent)	Times New Roman
Body size	12 pt
Line spacing	1.5 (one-and-a-half)
Paragraph alignment	Justified
Chapter headings	"CHAPTER n" + title, bold, centered, larger size (≈14–16 pt)
Section headings (1.1, 1.2 …)	Bold, left-aligned, numbered by chapter
Sub-section headings (1.1.1 …)	Bold, left-aligned

3. Page Numbering

- Preliminary pages (Acknowledgement, Abstract, Contents, List of Tables, List of Figures, Abbreviations): lower-case Roman numerals — i, ii, iii …
- Main report (Chapter 1 onward through References): Arabic numerals starting at 1, continuing through the References section.
- Title page and Certificate page carry no visible page number.

4. Document Order (front matter → body)

Students may add extra chapters as needed based on their base paper - not limited to 5.

1. Title page (see No.5)
2. Certificate page
3. Acknowledgement
4. Abstract
5. Contents (Table of Contents)
6. List of Tables (only if the report has tables)
7. List of Figures (only if the report has figures)
8. Abbreviations (only if abbreviations/acronyms are used)
9. Chapter 1 – Introduction
10. Chapter 2 – Literature Survey

11. Chapter 3 – Proposed Methodology
12. Chapter 4 – Results and Discussions
13. Chapter 5 – Conclusion and Future Work
14. References
15. Appendix

5. Title Page Contents (in order)

- Title of seminar (bold, caps, large)
- "A SEMINAR REPORT submitted by"
- Name of student, Register No. (bold)
- "to the APJ Abdul Kalam Technological University in partial fulfillment of the requirement for the award of the Degree of Bachelor of Technology in Computer Science and Engineering"
- "Under the guidance of" + Guide's name and designation (Assistant/Associate Professor/Professor)
- College logo
- "Department of Computer Science and Engineering, Amal Jyothi College of Engineering (Autonomous), Kanjirappally-686518"
- Month and year (e.g. October 2026)

6. Certificate Page

- Standard certificate paragraph naming the seminar title, student name, register number, university, degree, branch, and academic year.
- Signature block: Internal Supervisor (guide) on the left; Seminar Coordinators (Dr. Anju J Prakash, Ms. Shiney Thomas) on the right; Head of Department (Dr. Juby Mathew) below, centered.

7. Chapter Content Guidelines

Chapter 1 – Introduction

1.1 Overview, 1.2 Motivation and Objectives, 1.3 Report Organization (summarize what each subsequent chapter covers).

Chapter 2 – Literature Survey

Discussion of prior/related work with appropriate sub-headings; every claim must carry a numbered citation, e.g. [1].

Chapter 3 – Proposed Methodology

Subsections as needed for the specific paper/system (methods, data collection, experimental setup, key topics, figures/tables).

Chapter 4 – Results and Discussions

Findings, summary of key findings, critical analysis; include result tables/figures.

Chapter 5 – Conclusion and Future Work

Summarize outcomes and possible extensions.

8. Tables and Figures

- Numbered by chapter: Table x.y, Figure x.y (e.g., Table 4.1, Figure 4.1).
- Caption placed above tables and below figures.
- Every table/figure must be listed in the List of Tables / List of Figures with page number.
- Every table/figure must be referred to in the body text before it appears.

9. Abbreviations Page

Two-column list: abbreviation on the left, full expansion on the right, alphabetically or order-of-first-use.

10. References

- Numbered in square brackets in order of citation: [1], [2], …
- Each entry: Author(s), "Title of paper," Journal/Conference name, volume/issue, year.
- In-text citations must match the reference list numbering exactly.

11. Binding Order

Binding style: Soft bind, 1 copy

Before the report is bound, attach after the report as Appendix (in this order):

1. The seminar PPT, printed 4 slides per page.
2. The Seminar Base Paper.

SAMPLE REPORT TEMPLATE

TITLE OF SEMINAR

A SEMINAR REPORT

submitted by

NAME OF STUDENT
REGISTER NO

to

the APJ Abdul Kalam Technological University
in partial fulfillment of the requirement for the award of the Degree

of

Bachelor of Technology

in

Computer Science and Engineering

Under the guidance of

Dr./Prof./Mr./Ms. NAME OF FACULTY
Assistant Professor/Associate Professor/Professor

AMAL JYOTHI
COLLEGE OF ENGINEERING
AUTONOMOUS
KANJIRAPPALLY

Department of Computer Science and Engineering
Amal Jyothi College of Engineering(Autonomous)
Kanjirappally-686518

October 2026

DEPT. OF COMPUTER SCIENCE AND ENGINEERING

AMAL JYOTHI COLLEGE OF ENGINEERING

(AUTONOMOUS)

KANJIRAPPALLY

CERTIFICATE

*This is to certify that the seminar report entitled " **Seminar Title** " submitted by **Student Name (Register No.)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering during the year 2026-2027, is a bonafide work carried out by him/her under my guidance and supervision.*

Dr./Prof./Mr./Ms. Guide Name	Dr. Anju J Prakash Ms. Shiney Thomas
Internal Supervisor	Seminar Coordinators

Dr.Juby Mathew

Head of Department, CSE

ACKNOWLEDGEMENT

First of all I sincerely thank the **Almighty GOD** who is most beneficent and merciful for giving me knowledge and courage to complete the seminar successfully.

I derive immense pleasure in expressing my sincere thanks to our Manager, **Rev. Fr. Boby Alex Mannamplackal**, to our Director(Administrator), **Fr. Dr. Roy Abraham Pazhayaparampil** and to our Principal, **Dr. Lillykutty Jacob** for their kind co-operation in all aspects of my seminar.

I express my gratitude to **Dr. Juby Mathew**, HOD, Department of Computer Science and Engineering his support during the entire course of this seminar work.

I express my sincere thanks to **Ms. XYZ**, my internal guide and **Dr. Anju J Prakash** and **Ms. Shiney Thomas**, our Seminar Co-ordinators for their encouragement and motivation during the seminar.

I also express my gratitude to all the teaching and non teaching staff of the college especially to our department for their encouragement and help during my seminar.

Finally I appreciate the patience and solid support of my parents and enthusiastic friends for their encouragement and moral support for this effort.

NAME OF STUDENT

ABSTRACT

Integrity is the practice of being honest and showing a consistent and uncompromising adherence to strong moral and ethical principles and values. In ethics, integrity is regarded as the honesty and truthfulness or accuracy of one's actions. Integrity can stand in opposition to hypocrisy, in that judging with the standards of integrity involves regarding internal consistency as a virtue, and suggests that parties holding within themselves apparently conflicting values should account for the discrepancy or alter their beliefs. The word integrity evolved from the Latin adjective integer, meaning whole or complete. In this context, integrity is the inner sense of "wholeness" deriving from qualities such as honesty and consistency of character. As such, one may judge that others "have integrity" to the extent that they act according to the values, beliefs and principles they claim to hold.

An individual's value system provides a framework within which the individual acts in ways which are consistent and expected. Integrity can be seen as the state or condition of having such a framework, and acting congruently within the given framework.One essential aspect of a consistent framework is its avoidance of any unwarranted (arbitrary) exceptions for a particular person or group—especially the person or group that holds the framework. In law, this principle of universal application requires that even those in positions of official power be subject to the same laws as pertain to their fellow citizens.

CONTENTS

LIST OF TABLES

LIST OF FIGURES

ABBREVATIONS

WAT	Wave Atom Transformation
LPQ	Local Phase Quantization
CNN	Convolutional Neural Network
LBP	Local Binary Pattern
PCB	Printed Circuit Board
BSIF	Binarized Statistical Image Features
LivDet	Liveness Detection
VGG	Visual Geometry Group

CHAPTER 1

INTRODUCTION

1.1 Overview

Biometric security is a mechanism which is used to authenticate and ....

1.2 Motivation and Objectives

..To detect these fingerprint spoofing .... etc.

1.3 Report Organization

This section gives the organization of the report work. Overall, this report work gives a clear and thorough study about the existing study of the proposed system and gives a description about the implementation details. The second chapter gives a detailed description about the literature survey. Third chapter gives a brief idea about the implementation details including hardware and software requirements, outline of the proposed methodology. Fourth chapter depicts the Results and Discussions and fifth Chapter gives the Conclusion and Future Work. The Appendix shows the Screen Shots and Sample Code of the proposed systems.

CHAPTER 2

LITERATURE SURVEY

2.1 Discussion of earlier/older works

The Fingerprint Recognition System has ubiquitous deployment in many day-to-day applications, such as financial transactions, international border security, unlocking a smart phone[1]

Introduction The Fingerprint Recognition System has ubiquitous deployment in many day-to-day applications, such as financial transactions,[4] international border security, unlocking a smart phone, etc.

2.2 Create appropriate subheadings

This paper proposes a method to obtain the discriminant features for fingerprints whether they are real or spoofed, by performing feature level extractions [10]. The fingerprint image is preprocessed using five preprocessing operations. The preprocessing operations were selected which have the lowest validation errors.

1. Image reduction with different ratios Images are reduced using “Bilinear Interpolation”. They are used to verify the effect of resolution on the classifier performance.

2. Region Of Interest (ROI) Many fingerprints from some datasets are not centered; background represents large part of image. To input the classification system the

largest area that comprises foreground fingerprints, a simple ROI method is used.

They are used to verify the effect of resolution on the classifier performance.They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance. They are used to verify the effect of resolution on the classifier performance.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Add required subtopics as per paper

3.1.1 Xyz

Research Methods: Description of the methods and techniques used in the paper.

Data Collection: How data was gathered, if applicable.

Experimental Setup: Details of any experiments or simulations conducted.

Key Topics: Detailed discussion of the main topics covered in the paper.

Figures/Tables: Include and describe any figures, tables, or charts used in the paper for better understanding.

MATLAB supports graphical user interface (GUI) features for developing applications. For graphically designing these GUIs, MATLAB includes a GUIDE (GUI development environment) which also..

3.2 Xy Z

3.2.1 Xy Z

From these, only Digital-Persona was selected for the training and testing process in order to reduce the time required for processing. It consists of 1000 live images in both training as well as testing sets whereas fake images are obtained from four materials ie. Ecoflex

3.2.2 Xy Z

Wave Atom Transformation for Image Enhancement The fingerprint images obtained from the LivDet 2015 datasets may contain some noise.... and assigned it with two values 1 and 2. If the predicted value is 1, then it is considered as LIVE else it is FAKE.

CHAPTER 4

RESULTS AND DISCUSSIONS

This section gives as a detailed analysis of the proposed approach. Findings: Summary of the key findings or results presented.

Summary of Key Findings: Major takeaways from the seminar.

Analysis: Critical analysis of the information presented, including insights and implications.

We evaluated the performance on three scenarios to compare which one provides better performance.

Evaluation of CNN with feature vector as input In this setting, the fingerprint images in the dataset are firstly enhanced using Wave Atom Transformation method in order to remove the unwanted noise from the image pixels..... It took only 00:00:20 hrs at the time of testing whereas other two methods took more time. Figure 4.1 shows the gradient value obtained in each case. The performance Table 4.1 depicts that Ecoflex 00-50

Table 4.1: Correct Rate and Error Rate obtained for CNN with feature vector as input

Materials Used (Live and Fake)	CR	ER
Ecoflex 00-50	0.788	0.211
Gelatine	0.808	0.192
Latex	0.802	0.198
WoodGlue	0.808	0.191

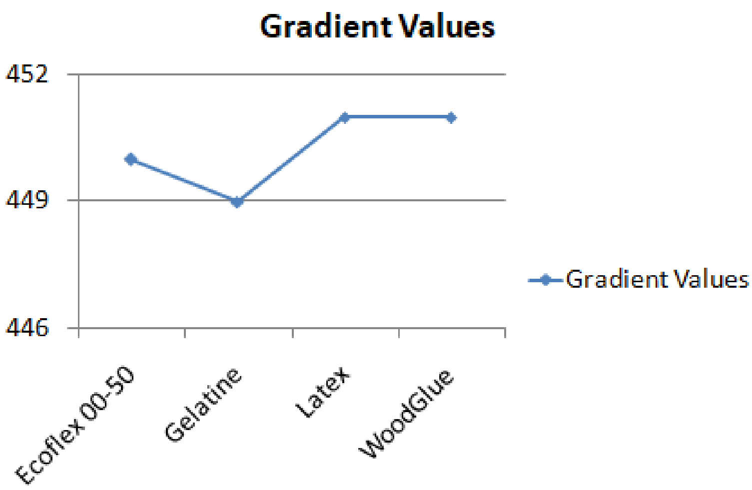

Figure 4.1: Graphical representation of gradient value obtained by each materials in the (ii) scenario

CHAPTER 5

CONCLUSION AND FUTURE WORK

The main limitation of many of the existing anti-spoof methods is the poor quality of the datasets obtained. The spoofing materials of poor image quality decreases the overall performance of the system. So we need either a best hardware for image acquisition or a good image processing method to obtain better results , but most of the hardware are expensive.

Here we proposed a method ..... The proposed WAT enhanced method yields an average accuracy of 85.8%. The WAT enhanced image with minutiae features extracted, achieved an average accuracy of 79.8%. We also computed the performance of CNN without any enhancement on image as input and obtained an average accuracy of 91.6%.

REFERENCES

[1] **Tarang Chugh, Kai Cao, and Anil K. Jain**, "Fingerprint Spoof Buster: Use of Minutiae-centered Patches," IEEE Transactions On Information Forensics And Security, 2018.

[2] **Manisha Redhu and Dr.Balkisha**, "Fingerprint Recognition Using Minutiae Extractor," International Journal of Engineering Research and Applications (IJERA),Vol. 3, Issue 4, Jul-Aug 2013.

[3] **Emanuela Marasco and Arun Ross**, "A Survey on Anti-spoofing Schemes for Fingerprint Recognition Systems," ACM Comput. Surv.47, 2, Article 28, Nov 2014.

[4] **Kumar Abhishek, Ashok Yogi**, "A Minutiae Count Based Method for Fake Fingerprint Detection," Second International Symposium on Computer Vision and the Internet (VisionNet'15), 2015.

[5] **Joshua J. Engelsma, Kai Cao, and Anil K. Jain**, "RaspiReader: An Open Source Fingerprint Reader Facilitating Spoof Detection," Life Fellow, IEEE, arXiv:1708.07887v1 [cs.CV], 25 Aug 2017.

[6] **Athos Antonelli, Raffaele Cappelli, Dario Maio, and Davide Maltoni**, "Fake Finger Detection by Skin Distortion Analysis," IEEE Transactions on Information Forensics and Security, Vol. 1, No. 3, Sep, 2006.

[7] **Rodrigo Frassetto Nogueira1, Roberto de Alencar Lotufo,and Rubens Campos Machado**, "Fingerprint Liveness Detection using Convolutional Neural Networks," IEEE Transactions on Information Forensics and Security, 1556-6013 (c) 2015.

[8] **Diego Gragnaniello, Giovanni Poggi, and Carlo Sansone and Luisa Verdoliva**, "Fingerprint Liveness Detection based on Weber Local Image Descriptor," IEEE Journal, 2013.

[9] **Luca Ghiani, Gian Luca Marcialis, and Fabio Roli**, "Fingerprint Liveness Detection by Local Phase Quantization," 21st International Conference on Pattern Recognition(ICPR 2012), Nov 2012.

[10] **Samruddhi S. Kulkarni and Dr. Hemprasad Y. Patil** , "A Fingerprint Spoofing Detection System Using LBP," International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT), 2016.

While binding, after this report add PPT (printed four slides per page. Then add the Seminar Base Paper.